Multi-Scale Probabilistic and Anomaly-Aware Decision Framework for Early Cellular Pathology and Transparent Clinical Reasoning

Postdoctoral Fellowship Research Proposal Summary

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Abstract

Modern medical diagnostics suffer from two critical bottlenecks: the *early detection gap* (pathologies originate at the cellular level long before macro-symptoms appear) and the *black-box dilemma* (existing AI models lack uncertainty quantification and transparent causal reasoning). This project proposes a rigorous, two-tier computational architecture that bridges unsupervised cellular-level anomaly detection with macro-level Dynamic Bayesian Networks (DBNs). By translating quantitative statistical methods traditionally used in high-energy physics (such as profile likelihoods, density estimation, and Monte Carlo sampling) into biomedical workflows, this framework delivers an open-source, uncertainty-quantified clinical decision support tool designed to assist primary care physicians and empower vulnerable populations.

Scientific Rationale & Objectives

The overarching goal is to construct a modular, transparent, and mathematically rigorous diagnostic pipeline spanning micro-scale signal detection and macro-scale clinical inference.

Tier 1: Micro-Scale Cellular Anomaly Detection

Pathological shifts initiate as subtle biochemical or spatial deviations at the cellular or micro-environmental level.

- **Data Ingestion & Telemetry:** Modeling high-dimensional biological telemetry (e.g., microbial chemical markers, smart probiotic sensor outputs, and multi-modal medical imaging).
- **Unsupervised Representation Learning:** Deployment of Variational Autoencoders (VAEs), Normalizing Flows, and Quantile Regression to model healthy baseline distributions.
- **Probabilistic Density Bounds:** Establishing smooth, patient-specific uncertainty envelopes rather than rigid binary classification thresholds to minimize false positives.

Tier 2: Macro-Scale Clinical Causal Reasoning

To translate raw anomaly scores into actionable clinical insights without relying on opaque black-box classifiers:

- **Causal Directed Acyclic Graphs (DAGs):** Encoding established pathophysiological pathways, symptom interactions, and disease progression models.
- **Dynamic Bayesian Networks & MCMC:** Implementing Markov Chain Monte Carlo (MCMC) sampling routines to handle biological feedback loops and longitudinal patient updates.
- **Active Diagnostic Testing:** Integrating Expected Information Gain (EIG) metrics to dynamically recommend the most informative, cost-effective diagnostic test for clinicians to run next.

Methodological Architecture & Integration Pipeline

The framework enforces a strict architectural decoupling between low-level unsupervised density estimation and high-level causal inference:

Data Layer: Multi-modal Telemetry / Cellular Sensors / Medical Imaging
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Tier 1: Unsupervised Anomaly Detection (VAEs, Normalizing Flows, Quantile Regression)
⇓ *(Continuous Anomaly Scores & OOD Density Metrics)*
Tier 2: Causal Inference Engine (Dynamic Bayesian Networks, MCMC Sampling, Feedback Loops)
⇓
Output: Transparent Posterior Probabilities & Uncertainty-Quantified Clinical Decision Support

Rigorous Uncertainty Quantification

Drawing from extensive experience in high-energy physics statistical profiling (e.g., profile likelihood ratio fits and systematic error tracking), this project explicitly separates *aleatoric uncertainty* (inherent biological noise) from *epistemic uncertainty* (model parameter ambiguity). This ensures clinicians receive transparent confidence intervals alongside every diagnostic hypothesis.

Feasibility, Open-Source PoC, and Host Synergy

Existing Open-Source Proof of Concept

The foundational architecture for these systems has already been prototyped and open-sourced by the applicant:

- `AnomalyDetection-Medical`: Modular unsupervised anomaly detection pipelines for high-dimensional medical distributions.
- `Medical-Diagnosis-Engine`: Probabilistic causal inference engine structured for joint disease posteriors and uncertainty quantification.

Transferability of Quantitative Rigor

Transitioning advanced statistical and computational techniques from experimental high-energy physics (CMS Collaboration, CERN) to biomedical data science provides a unique methodological advantage. The rigorous handling of massive background noise, multi-variable parameter profiling, and automated pipeline orchestration directly addresses the core reliability challenges facing modern medical AI.

Broader Impact

This project rejects the over-commercialization and opacity of closed-source medical AI. By delivering an open-source, auditable clinical decision support tool, the framework directly assists compassionate primary care physicians in providing high-quality, affordable baseline healthcare to underserved communities.